8

How Tournaments Impact Decisions to Vote

Thus far, this book has offered evidence that LDP politicians use tournaments to get elected. Before election campaigns, they dangle the promise of central government resources over the heads of voters in their electoral districts. After votes are counted, they calculate how much support they received from each of the municipalities therein. Then, via pressure exerted on central government bureaucrats, they engineer a situation in which the largest prize goes to the most supportive municipality, the second-largest prize goes to the second-most supportive municipality, and so on. Making the difference in amounts received by the most supportive and second-most supportive municipalities larger than the difference in amounts received by the second-most and third-most supportive, which is in turn larger than the difference in amounts received by the third-most and fourth-most supportive, and so on, creates a situation where a handful of votes in a highly supportive municipality can be enough to bring about meaningful change in a municipality's resource allocation. This gives voters in those municipalities reason to set aside whatever personal feelings they harbor toward the incumbent and reason that, given she will very likely be reelected, they may as well use their vote to make sure their municipality continues to be prioritized when it comes to resource allocations. For the incumbent, the competition this engenders among voters in supportive municipalities is a convenient way of amassing enough votes to win.

In Chapters 6 and 7, I looked for evidence of tournaments in postelection allocations of national treasury disbursements (NTD). Headline findings were threefold. One, within electoral districts, the amount of NTD municipalities receive in the years following Lower House elections was found to be a function of its position in a rank order of municipalities, constructed on the basis of support for the LDP incumbent. More specifically, municipalities at higher ranks (exhibiting greater support for their LDP incumbent) received larger postelection resource allocations

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

(Hypothesis 1). Two, also within districts, the amount of NTD municipalities receive was found to increase with their position in this ranking, but not in a linear fashion, in a *convex* fashion. By this, I mean that the returns to increases in rank increase at an increasing rate as a municipality climbs the ranks. A municipality that increases its rank from second to first place can expect a much larger increase in its NTD allocation than a municipality that experiences an equivalent one-rank increase at a lower position in the ranking (such as from six to five). This is evidence that LDP incumbents deliberately make the amounts available for municipalities to fight over larger at higher ranks (Hypothesis 2).

Three, across electoral districts, the total amount of NTD received by a district's voters is found to vary according to the relative size of the municipalities therein. The more similarly sized the municipalities in a given electoral district are, the less (money) is needed to motivate them to compete against each other. Thus, when the amount of NTD received by all municipalities in each district is tallied up, the total amount of money electoral districts receive is found to increase with the degree of asymmetry in municipality size (Hypothesis 3). Even though districts with asymmetrically sized municipalities receive more money, my analyses also reveal that they tend to deliver less support for the LDP (Hypothesis 4).

In this chapter, I look for evidence of a tournament in voter decisions to turn out and vote. The theory expects that decisions to vote will hinge on *where* in the ranking a municipality is expected to place. Why? Making a municipality's resource allocation contingent upon its rank, and designing rewards so that they increase with rank at an increasing rate creates a situation in which highly supportive municipalities will all vie for first place. This creates a situation in which a handful of votes could be enough to change a municipality's ranking and hence bring about a sizeable increase in its resource allocation. Thus, in municipalities projected to place well in the tournament, a single vote wields considerable influence over the amount of resources a municipality receives. All else equal, then, we can expect that voters will be systematically more motivated to go to the polls in municipalities projected to place highly in the tournament, which are in the running for a very large prize. In these municipalities, voters might realize that their vote is immaterial to the *outcome* of the election (after all, the incumbent is likely to win), but material to the amount of money their municipality receives after the election. Voters turn out to avoid being the reason their municipality misses out on a large prize, which would be the outcome if it failed to main the high ranking it attained last time or missed out on boosting its ranking this time.

The situation is different in municipalities not projected to place highly. These municipalities may not regularly deliver large shares of their

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

voting population to the LDP incumbent and may have pockets of support for *non*-LDP candidates. Amongst these relatively unsupportive municipalities, increases in rank will still bring about increases in resources, but any increase in resources that comes from an increase in rank at the middle-to-lower end of the ranking will be smaller. Because of this, neck-and-neck competition at the middle-to-lower end of the ranking will be much less likely to materialize. Amongst supportive municipalities, it is the *anticipation* of a large increase in resources that galvanizes voters to make the effort to go to the polls, regardless of how they feel about the incumbent and regardless of whether they think their vote will affect the election's outcome. As more and more voters use their vote this way, neck-and-neck competition for rank increases materializes, which further increases the influence of a single vote. Among less supportive municipalities, the situation is different: knowing that increases in rank are unlikely to yield anything short of a modest increase in resources, more voters will calculate that they may as well stay home on election day, given that the resources their municipality stands to win will hardly be affected by whether or not they personally vote. As more and more voters make this calculation, turnout will decrease, which in turn means that more votes become necessary to change the municipality's ranking.

In this chapter, I devise three sets of tests that examine whether the motivation to vote differs according to a municipality's projected position in the ranking. To conduct these, I operationalize new variables that capture the degree to which a municipality concentrates its votes on a single LDP winner.[1] Using these variables, I first examine whether, within electoral districts in the same election, turnout increases as municipalities concentrate larger shares of valid votes cast on a single LDP winner. This is Hypothesis 5. Next, I continue to restrict the comparison to municipalities in the same district in the same election and examine whether, if I rank municipalities in terms of the degree to which votes are concentrated on a single LDP winner, increases in vote concentration at the top of the ranking, amongst municipalities that are already concentrating many votes on a single LDP winner, bring about larger increases in turnout than increases in vote concentration at the middle or bottom of the ranking. This is Hypothesis 6. As with the tests in Chapters 6 and 7, all these tests take confounders seriously, pit the predictions of the tournament theory against those of rival theories, and are sensitive to the specifics of the electoral system under which incumbents are competing.

[1] The support variables operationalized in Chapter 6 are not used for these tests because the denominator in those support variables (voting population) is the same as in turnout (voting population).

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

By comparing turnout rates between municipalities in the same district-year, these first sets of tests hold constant the effect of all variables influencing turnout that operate at the level of the district-year. A factor influencing turnout at this level, which we know exercises a decisive influence on voter decisions to go to the polls, is the *competitiveness* of the race. A vast literature in political science has established that when a race is projected to be competitive, more voters will go to the polls. When a race is not projected to be competitive, in contrast, more voters decide to sit the election out. By definition, competitive elections mean narrow margins of victory, and narrow margins of victory increase the probability that one's vote might *matter*, in the sense of helping to determine who wins.

My third test leverages variation across electoral districts in the degree to which races are projected to be competitive to conduct an additional test of Hypothesis 5. To ascertain whether the motivation to vote differs systematically according to *where* in the ranking a voter's municipality is likely to end up, I examine whether projections of increased competitiveness have a *smaller* impact on turnout rates in municipalities projected to place highly in the ranking. In municipalities projected to place highly, the theory expects that motivations to vote will be driven primarily by voters' desire to realize a larger resource allocation. Thus, turnout in highly supportive municipalities should be *less* affected by situational elements of a given election, such as closeness. It is likely to remain high even when an election is not projected to be close. In contrast, in municipalities not projected to place highly, turnout is more likely to be affected by situational elements, such as competitiveness.[2] Thus, we can expect that turnout in the less supportive municipalities within a given electoral district will be more sensitive to projections of closeness. Like my other tests, the tests crafted in this section consider whether my results are better explained by rival theories, particularly one positing that turnout in Japanese elections is a function of a conditional relationship between district-level competitiveness and population density (Cox, Rosenbluth and Thies 1998).

8.1 EMPIRICAL STRATEGY

Chapter 4 fleshed out the theoretical underpinnings of the hypotheses this book tests. For ease of access, here are the two examined in this chapter:

- **Hypothesis 5**: Within electoral districts, turnout rates are a function of a municipality's position in a rank ordering of municipalities,

[2] I use competitiveness and closeness interchangeably here.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

constructed on the basis of support for the LDP winner in the most recent election.

- **Hypothesis 6**: Within electoral districts, the impact of increases in rank on turnout in a municipality increases as municipalities climb the ranks.

8.1.1 Variables

As I explained in Chapter 6, my data consist of yearly observations (where t indicates the year) of electoral districts (d), municipalities (m), and candidates (c). Let $n_{d,t}$ capture the number of municipalities in district d in year t.[3] First, I created variables that capture the degree to which municipality m favored a single LDP winner in an election held in year t. Let $v_{m,t}$ represent the number of valid votes cast in municipality m at time t. This is available for $t \in E = \{e_1, e_2, ...\} = \{1980, 1983, ...\}$, the years in which Lower House elections were held. Let $v_{c,m,t}$ represent the number of votes cast for candidate c in municipality m at time t. Let $LDP_{c,t}$ indicate whether candidate c was a member of the LDP at time t and $w_{c,t}$ indicate whether candidate c won a seat in district d at time t. I calculated:

$$\text{Single LDP Winner VS}_{m,t} = \frac{max_{c \in m}\{LDP_{c,t} w_{c,t} v_{c,m,t}\}}{v_{m,t}} \tag{8.1}$$

In words, this takes the number of votes received by all LDP winners in municipality m at time t, identifies the LDP winner who won the largest number of votes, and divides the number of votes won by this LDP winner by the number of valid votes cast in the municipality.[4] To obtain a high value on Single LDP Winner $\text{VS}_{m,t}$, municipalities have to cast more of their votes, however many are cast, for a single LDP winner. Note that municipalities in electoral districts without a single LDP winner all receive 0 on Single LDP Winner $\text{VS}_{m,t}$.

In the preelectoral reform period (1980–1993), candidates in Lower House elections competed in multimember districts (MMDs). In the post-electoral reform period, they competed in single-member districts. Thus, the number of winning candidates (and hence, the number of LDP winners) varies across district-years. A municipality that manages to send

[3] As with the analyses in Chapter 6, municipalities straddling more than one electoral district are excluded from the creation of these variables.

[4] The number of valid votes cast is calculated by taking the sum of the number of valid votes cast for all candidates who ran in the district-year.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

40% of votes cast to a single LDP winner in a district that produces four LDP winners could be said to have more strongly identified with that winner than a municipality that sent the same 40% of votes cast to a single LDP winner in a district with only one LDP winner. To make sure that district-level differences in the number of winners (district magnitude) are not producing systematic differences in Single LDP Winner $VS_{m,t}$ that could have consequences for my results, I also constructed a ranked version of this variable, which is comparable across municipalities with different numbers of LDP winners.

Concretely, for the n municipalities located in district-year dt, I ranked them by values on Single LDP Winner $VS_{m,t}$ and then standardized the ranking so that the municipality with the smallest Single LDP Winner $VS_{m,t}$ (the lowest degree of identification with a single LDP winner) always receives 0 on Rank(Single LDP Winner $VS_{m,t}$) and the municipality with the largest Single LDP Winner $VS_{m,t}$ (highest degree of identification with a single LDP winner) always receives 1 on Rank(Single LDP Winner $VS_{m,t}$).[5] Note that as municipalities in districts without LDP winners all have 0 on Single LDP Winner $VS_{m,t}$, they cannot be ranked.[6]

Given that the variables operationalized in Chapter 6 were explicitly designed to capture the municipality's position in the ranking, which is also what I want to capture here, why can I not use these variables here? Using these support variables (Best LDP $VS_{m,t}$ and Sum LDP $VS_{m,t}$ and their corresponding ranked versions) would not produce compelling tests of my hypotheses on account of the fact that the denominator in all four support variables is the number of eligible voters in municipality m, which is also the denominator in turnout (as I explain below). Thus, by construction, municipalities with higher values on the four support variables used in Chapter 6 will have higher values on turnout. In contrast, a high value on the variable operationalized above, Single LDP Winner $VS_{m,t}$, simply means that municipality m sent most votes cast to a single LDP winner; it is agnostic about whether the number of votes cast was high

[5] To do this, I first ranked municipalities so that the lowest-ranked municipality receives 0 and the highest-ranked municipality receives one less than the number of municipalities in the district ($n_{d,t} - 1$). Then, dividing each municipality's value on this ranking by the total number of municipalities in its district-year minus 1 ($n_{d,t} - 1$) standardizes the index so that in each district-year, the municipality with the lowest degree of identification with a single LDP winner receives 0 and the municipality with the highest degree of identification with a single LDP winner receives 1.

[6] These municipalities receive NA on Rank(Single LDP Winner $VS_{m,t}$).

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

or low, relative to voting population. This variable is better suited to discerning whether increases in the degree to which votes are concentrated on a single LDP winner are associated with more voters making the decision to turn out.

Next, let $p_{m,t}$ represent the number of eligible voters in municipality m at time t (the voting population). My dependent variable, turnout in municipality m, is calculated by taking the number of valid votes cast in the municipality and dividing this by the number of eligible voters. More formally:

$$\text{Turnout}_{m,t} = \frac{v_{m,t}}{p_{m,t}}. \tag{8.2}$$

Finally, as a measure of the competitiveness of the race in a given electoral district in a given year, I compute $\text{Margin}_{d,t}$. This is the number of votes cast for the last winner in district d at the election at t, minus the number of votes cast for the first loser, divided by the total number of valid votes cast in district dt. Analyses of races with a single winner routinely use the difference in number of votes won between the winner and the first loser (or first runner up), divided by votes cast, as an indicator of the competitiveness of a district. Studies of Japanese politics have shown that the same operationalization, but between the "last" winner and the first loser, is appropriate when there is more than one winner (e.g., Ariga 2015). When there are four winners, then, $\text{Margin}_{d,t}$ is the difference between the number of votes won by the fourth-placed candidate (the last winner) and the number of votes won by the fifth-placed candidate (the first loser). Generally speaking, larger values of $\text{Margin}_{d,t}$ mean a larger gap in votes received by the last winner and the first loser, which indicates a *less* competitive race.

8.1.2 Potential Confounders

My goal is to ascertain whether a municipality's projected position in a rank order of municipalities in its district, constructed on the basis of support for the LDP incumbent, exercises an independent effect on voter decisions to turn out. More specifically, I want to know if increases in position in the ranking bring about increases in turnout, and if those increases bring about even larger increases in turnout when municipalities are at the top of the ranking than when they are at the middle or bottom. To conduct compelling tests of this hypothesis, I must be mindful of alternative variables that could influence the relationship between vote concentration and turnout, and I must control for these variables.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

The political science literature on the drivers of turnout is vast.[7] Studies of turnout in Japanese elections have established the influence of the following six variables: the projected closeness of the race (Jou 2009; Cox, Rosenbluth and Thies 1998), the electoral system (Smith 2018*b*), changes in electoral rules (Kohno 1997*b*), the number of seats available in the district (district magnitude) (Muraoka and Barceló 2019), the number of representatives elected in the district per voter (Fujimura 2020; Horiuchi 2005), and urbanness (Cox, Rosenbluth and Thies 1998; Richardson 1973), respectively. Let us briefly review the findings on each.

First, a vast literature establishes that when elections are projected to be *close*, meaning the margin of victory between the winner and loser is expected to be narrow, more voters make the effort to go to the polls (e.g., Blais 2000; Endersby, Galatas and Rackaway 2002; Jou 2010).[8] One explanation for this is rooted in voter psychology and specifically, in the desire for one's vote to matter. This explanation holds that the closer an election is projected to be, the higher the chance that a single vote could decide the outcome. Because going to the polls is costly and takes time away from other activities, voters will be more willing to incur these costs when they believe their vote might influence the outcome. Another explanation is rooted in the calculations of political elites (e.g., Key 1949; Cox and Munger 1989). This locates the impetus of decisions to turn out in the behavior of elites and argues that elites try harder to mobilize votes for their preferred candidates when elections are expected to be close, and this increased mobilization drives the positive relationship between closeness and turnout.

In the Japanese context, at least two studies have found evidence that district-level competitiveness influences turnout. Jou (2010) was interested in whether, under Japan's mixed-member majoritarian (MMM)

[7] In their meta-analysis of this literature, Cancela and Geys (2016) note three classes of explanatory variables that scholars have invoked to account for differences in turnout rates across geographically defined units. One is socioeconomic variables such as population, population density, or ethnic homogeneity; another is political variables such as closeness or political fragmentation; and the third is institutional variables such as electoral system, voter registration, or the presence of concurrent elections (see also Kostelka et al. 2023). Importantly, institutional variables are found to have an impact up to four times' as great as socioeconomic or political variables.

[8] After Downs (1957)'s observation that it is not rational for anyone to vote because the costs of voting vastly outweigh the likelihood that a single vote will ever be pivotal, Riker and Ordeshook (1968) argued that people vote for other reasons, such as to affirm their commitment to the democratic process or fulfil what they perceive as a social obligation. Dowding (2005) offers a summary. Despite the fact that a single vote is hardly ever pivotal, a large literature has nevertheless found that turnout increases in close elections. Thus, the desire to influence the outcome of elections appears to exercise at least some sway over individual decisions to turn out.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

electoral system, the addition of the proportional representation (PR) tier attenuated the relationship between competitiveness at the level of the single-seat district (SSD) and turnout. His reasoning was that in a mixed system like Japan's, voters who support parties with little chance of placing first in the SSD can influence those parties' seat tallies by turning out and casting a PR ballot for them. Because voters cast ballots for the SSD race and the PR race during the same trip to a polling station, the influence of the PR ballot could attenuate any relationship between SSD-level competitiveness and turnout. Notwithstanding this logic, his analyses of the 2005 and 2009 Lower House elections revealed that SSD-level competitiveness "emerges as having the most substantial impact on turnout among all socio-demographic and political variables tested" (Jou 2010, 1055).

Another study examined whether district-level competitiveness influenced turnout in Japanese elections during the period prior to electoral reform, when MMDs were used (Cox, Rosenbluth and Thies 1998). While the authors found evidence of this, they further hypothesized that the impact of district-level competitiveness depended on the density of social ties within districts. Their reasoning was that close elections produce higher turnout via the actions of elites, who engage in calculations as to whether or not efforts to mobilize votes will bear fruit. In districts with dense social ties ("high social capital"), they reasoned that elites understood that every additional voter contacted and urged to vote would have a larger number of friends and acquaintances to whom this message would be passed on. Thus, close elections would produce more intense elite mobilization, and hence, higher turnout, in districts with dense social ties. In support of this, they studied turnout at the district level in Lower House elections between 1967 and 1993. Controlling for differences between election years, they showed that less competitive elections (larger margins separating the last winner from the first loser) have lower turnout, but less competitive elections have even lower turnout in districts where social ties are dense (sparsely populated districts and districts with more women and children under fifteen). In urban districts, where social ties are the least dense, the effect of closeness on turnout was found to be "almost nil" (Cox, Rosenbluth and Thies 1998, 468).

Second, another strand of literature holds that turnout is decisively influenced by *electoral system*. Specifically, elections under PR have been shown to have higher turnout rates than elections under majoritarian systems. Why? As I alluded to in my discussion of Jou (2010), under PR, seats accrue to parties in proportion to their vote share; thus, there are few wasted votes. In a majoritarian system, in contrast, votes cast for losing candidates are wasted, as are votes cast for district winners in excess

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

of the number needed to win. The larger number of wasted votes in a majoritarian system is thought to depress turnout. Cox, Fiva and Smith (2016), however, caution against blanket statements that turnout will be higher under PR. They point out that the winner-take-all nature of elections under majoritarian rules means that close races are likely to produce turnout levels that exceed those in a PR system, but uncompetitive races will not. Thus, the question of whether turnout is higher under PR or majoritarian rules boils down to inquiring as to the share of races in the majoritarian system that are competitive. If there are few competitive races, then aggregate turnout will be low and will increase if the system switches to PR. If there are many competitive races, in contrast, aggregate turnout is likely to be high and may suffer a decline if the system is switched to PR. In a comparison of turnout in the last three elections held under SNTV-MMD and the first three under MMM, Smith (2018*b*) finds that turnout is lower under the new system.

Third, Kohno (1997*b*) studied the impact of *electoral reforms* on turnout. His analyses showed that turnout declined in the first election held under Japan's new mixed-member system (in 1996) and also fell in the first election held under SNTV-MMD (in 1947). He argues that these declines can be attributed to the complexity and/or newness of the system, and/or the fact that the redrawing of district boundaries meant that many voters faced entirely new slates of candidates, with which it was costly to familiarize oneself.

Fourth, turnout is also thought to be decisively influenced by *district magnitude* (the number of representatives elected in a district). Canonical models of electoral systems compare majoritarian systems, assumed to have a district magnitude of 1, with PR systems, assumed to have a district magnitude much greater than 1 (and sometimes, in the hundreds). But electoral systems qualifying as majoritarian can have district magnitudes greater than 1, while electoral systems qualifying as proportional can have smaller district magnitudes. Thus, more recent work seeks to isolate the independent effect of district magnitude itself. On the one hand, larger district magnitudes are thought to produce higher turnout on the grounds that an increase in the number of candidates running means that more voters would find a candidate aligned with their preferences. On the other hand, when district magnitudes are very large, the large number of candidates competing places a large cognitive burden on voters, who must acquire the information needed to adjudicate between them (Taagepera, Selb, and Grofman 2014). This is particularly the case when candidates rely on personal characteristics, as opposed to party labels, to distinguish themselves from co-partisans (Muraoka and Barceló 2019). As a result, district magnitudes over four are typically thought to produce

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

lower turnout. Muraoka and Barceló (2019) study this in Japan. They leverage the fact that until 2011, district magnitudes in municipal assembly elections increased discontinuously at certain population thresholds and was strictly applied (municipalities whose populations fell below a threshold had to call for new elections).[9] Among municipalities in their sample, whose district magnitudes ranged from 6 to 46, they found that increases in district magnitude was associated with a decline in turnout.

Fifth, the *number of representatives per voter* (electorate size) is also thought to influence turnout. In smaller electorates, representatives are elected with a smaller number of votes, which means that votes count more. Horiuchi (2005) conducts a cross-national study of this hypothesis, which includes extensive analyses with data from Japan. He finds that turnout is higher in municipal assembly elections than in national elections in Japan, and he argues that this is primarily (although, not only) due to the smaller electorate size. Across municipalities, he also finds that turnout in assembly elections is higher in smaller municipalities than in larger ones. Due to the way numbers of assembly members are assigned, smaller municipalities have a larger number of elected representatives per capita than larger municipalities. In his analyses of turnout in the 2005 and 2009 Lower House elections in Japan, Jou (2010) includes a control for electorate size. He explains that while Japan's 1994 electoral reform reduced the disparity across districts in terms of number of votes needed to elect a representative, it did not eliminate it. He counts several districts that have more than twice as many voters as the smallest district. Nevertheless, he found that this variable had no systematic relationship with turnout in the elections he examined. In contrast, a study of turnout in Upper House elections found that electorate size is negatively correlated with turnout (Fujimura 2020).

Sixth, early work on turnout in Japanese elections found that it was almost always higher in *rural communities* than in urban ones (Richardson 1973). The main reason for this, alluded to above when I discussed Cox, Rosenbluth and Thies (1998), is thought to be that voters in rural areas are more likely to be longstanding residents and therefore better integrated into networks and associations that could be harnessed by candidates for the purpose of vote mobilization. While candidates benefit from being able to tap into communities with well-established social networks everywhere, Japan's use of SNTV-MMD prior to 1994 made these networks of particular value to LDP candidates. Under this system, LDP candidates were campaigning against each other and had

[9] The Local Autonomy Law stipulated that if the number of municipal residents was less than or equal to 50,000, for example, the maximum number of council members could be 26.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

no party organization upon which to draw. They contested elections by emphasizing their personal characteristics, background, and connections, as well as what they planned to get done in the district. Between elections, they cultivated personal support organizations, called koenkai, by plying voters with all manner of favors, including assistance with requests, parties, and trips to the hot springs. It was koenkai members who did the bulk of the vote mobilization during the relatively short campaign period (McElwain 2008). If it is easier to mobilize votes in rural areas due to higher density of social ties, then we are likely to observe more vigorous vote mobilization by LDP candidates in those areas, and thus, higher turnout.[10]

The good news is that my research design controls for all these variables except the last. By including district-year fixed effects in the main tests of Hypotheses 5 and 6, I am able to hold constant the effects of all variables operating at this level. While these variables could be creating vast disparities in turnout rates across electoral districts, driving up turnout in some districts and driving it down in others, my use of district-year fixed effects enables me to look *within* district-years, to see if a municipality's projected position in the ranking exercises an independent effect on turnout, notwithstanding these other differences. Of the six variables, one operating at the level of the municipality, which I need to control for, is urbanness (or more broadly, density of social ties). I operationalize this with population density, proportion of the population employed in farming, and the proportion of population aged fifteen and under and sixty-five and older, respectively. Later in the chapter, I exclude district-year fixed effects from the specification, which enables me to leverage variation across districts in competitiveness to conduct an additional test of Hypothesis 5.

8.2 CONCENTRATING VOTES ON A SINGLE LDP WINNER INCREASES TURNOUT

First, across municipalities in the same electoral district in the same election, is the share of voters who bother to turn out and vote higher in municipalities that more strongly identify with a single LDP winner? To answer this question, Table 8.1 presents the results of two-way fixed effect

[10] Notwithstanding this logic, the analyses in Cox, Rosenbluth and Thies (1998) found that the relationship between ruralness and turnout at the district level in Lower House elections under SNTV-MMD is mediated by closeness, to the extent that there was almost no difference in turnout between rural and urban areas when races are uncompetitive. On the new system, there is some evidence that urban–rural differences in turnout have reduced (Jou 2010).

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

Table 8.1 *Municipalities that concentrate more votes on a single LDP winner have higher turnout in Lower House elections, 1980–2014.*

	Dependent variable: $\text{Turnout}_{m,t}$			
	(Model 1)	(Model 2)	(Model 3)	(Model 4)
Single LDP Winner $\text{VS}_{m,t}$	0.026***	0.019***		
	(0.002)	(0.003)		
Rank(Single LDP Winner $\text{VS}_{m,t}$)			0.006***	0.004***
			(0.001)	(0.001)
$\text{Turnout}_{m,t-1}$	0.727***	0.241***	0.732***	0.224***
	(0.008)	(0.011)	(0.008)	(0.012)
Fiscal $\text{Strength}_{m,t}$	−0.008***	0.009**	−0.008***	0.012***
	(0.001)	(0.003)	(0.001)	(0.003)
Dependent $\text{Population}_{m,t}$	0.136***	0.185***	0.144***	0.186***
	(0.010)	(0.015)	(0.010)	(0.017)
Farming $\text{Population}_{m,t}$	0.026***	0.051**	0.030***	0.045*
	(0.005)	(0.019)	(0.005)	(0.022)
Log($\text{Population}_{m,t}$)	−0.008***	−0.055***	−0.008***	−0.065***
	(0.0003)	(0.004)	(0.0003)	(0.005)
Log(Per Capita $\text{Income}_{m,t}$)	0.006***	0.015***	0.005**	0.015***
	(0.002)	(0.004)	(0.002)	(0.004)
Population $\text{Density}_{m,t}$	0.00004	0.022***	−0.00004	0.027***
	(0.0003)	(0.005)	(0.0003)	(0.005)
Observations	24,936	24,936	20,885	20,885
District-Year FE	Y	Y	Y	Y
Municipality FE	N	Y	N	Y
R^2	0.775	0.158	0.778	0.162

Robust standard errors clustered on municipality in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

regression models of $\text{Turnout}_{m,t}$ in the twelve Lower House elections held between 1980 and 2014 as a function of Single LDP Winner $\text{VS}_{m,t}$ (Models 1 and 2) and Rank(Single LDP Winner $\text{VS}_{m,t}$) (Models 3 and 4).[11] My sample is nonsplit municipalities in tournament-possible districts. All specifications control for turnout in the previous Lower House election, as well as six time-varying features of Japanese municipalities that could influence the relationship between vote concentration and turnout: population (log), per capita income (log), fiscal strength, population density, the proportion of the population employed in primary industries, and the proportion of the population aged fifteen and under

[11] Descriptive statistics pertaining to the variables used in these analyses appear in Table A.15.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

and sixty-five and over, respectively.[12] These are the same municipality-level variables controlled for in the analysis of transfers. My discussion above identified the latter three variables as potentially important confounders. I report robust standard errors clustered on municipality.

All specifications include district-year fixed effects, which control for features of district d at time t that could be increasing or decreasing turnout in all the municipalities therein. As I explained, a time-varying characteristic of a district-year thought to exert a positive impact on turnout is competitiveness: If a race is projected to be more competitive than usual, perhaps because of the entry of new candidates or a scandal involving an incumbent, voters may calculate that their vote has more of a chance of influencing the election's outcome, and thus, will be more likely to turn out. In addition to closeness, the above discussion identified electoral system, district magnitude, and number of representatives per voter as other district-specific variables that could influence turnout. By controlling for all features of district d at time t that could influence turnout, district-year fixed effects enables us to look *within* district-years. Thus, the coefficients on Single LDP Winner $\text{VS}_{m,t}$ and Rank(Single LDP Winner $\text{VS}_{m,t}$) will tell us whether municipalities that identified more strongly with a single LDP winner relative to other municipalities in the same district-year had higher turnout. In each specification, I report robust standard errors clustered on municipality.[13]

In Models 1 and 3, coefficients on my measures of vote concentration are positive and statistically significant. This indicates that within district-years, controlling for time-varying features of municipalities that matter for turnout, municipalities that concentrate more of the votes cast on a single LDP winner have higher turnout. Substantively, Model 1 of Table 8.1 tells us that a one-standard-deviation increase in Single LDP Winner $\text{VS}_{m,t}$ can be expected to increase turnout by 0.62%. As I explained in Chapter 6, this is the *average* effect of a one-standard-deviation increase, no matter where it takes place in the ranking. The theory expects that increases in voting concentration will exert a much larger impact on turnout when they occur at the top of the ranking than when they occur at the middle or bottom. The sample mean of $\text{Turnout}_{m,t}$ in these elections was 0.73, meaning that on average, 73% of eligible voters in a municipality turned out to vote. The coefficient on prior turnout is also positive and statistically significant. If voting patterns are persistent,

[12] Note that population density captures the number of thousands of people per square kilometer.

[13] The number of observations is smaller in Models 3 and 4 because municipalities are not ranked in terms of Single LDP Winner $\text{VS}_{m,t}$ when there are no LDP winners in a district.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

then prior turnout would have been influenced by prior vote concentration, which could account for the positive relationship between turnout in the previous election and turnout in this election. The persistence of voting patterns tends to diminish the size of our estimates of the effect of vote concentration. If I exclude prior turnout from Model 1, for example, the coefficient on Single LDP Winner $VS_{m,t}$ is more than three times' larger.

In Models 2 and 4, I add municipality fixed effects to the specifications in Models 1 and 3. A municipality fixed effect controls for time-invariant features of a municipality that could be influencing turnout, such as geographic features that influence the accessibility of polling stations. A municipality fixed effect takes care of the possibility that underlying differences of this nature between the municipalities in a district-year could be driving any observed relationship between vote concentration and turnout. Even after adding municipality fixed effects, I find that the coefficients on my vote concentration variables remain positive and statistically significant. This reduces concern that underlying differences between the municipalities are responsible for the observed effects of Single LDP Winner $VS_{m,t}$ and Rank(Single LDP Winner $VS_{m,t}$) in Models 1 and 3. What Models 2 and 4 tell us is that when the *same* municipality concentrates more of the votes cast on a single LDP winner, it has higher turnout than when it concentrates fewer votes.

It is likely that the lagged dependent variable (all models) and the municipality fixed effect (Models 2 and 4) are absorbing some of the effects of my control variables, particularly when they exhibit minimal variation over time. Nevertheless, controlling for other differences between municipalities, my results suggest that increases in fiscal strength, per capita income, the proportion of the population who are dependent, the proportion of the population employed in agriculture, and population density are all associated with higher turnout. On the other hand, increases in population per se is associated with lower turnout.

In sum, in Chapter 6 I found that within district-years, exhibiting high levels of support for a single LDP winner is associated with receiving more government resources after the election. Here, I have shown that within district-years, concentrating more votes on a single LDP winner is also associated with higher voter turnout. This is consistent with the idea that as municipalities become more and more devoted to the election of a single LDP winner, their chance of winning the largest prize increases, which motivates more voters in the municipality to turn out to vote. The fact that variation in turnout is a function of the degree to which a municipality identifies with a single LDP winner is further evidence that LDP politicians win elections by engaging in group-based clientelism (GBC).

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

8.2 *Concentrating Votes Increases Turnout*

8.2.1 *Before Electoral Reform, 1980–1993*

Next, I divide up the time period, which enables me to accomplish two goals. One is to examine whether turnout is a function of vote concentration under *both* electoral systems Japan has used in the 1980–2014 period. My analyses of resource allocations in Chapter 6 revealed that LDP politicians used tournaments in both periods, so I expect that turnout will also be a function of vote concentration in both periods. The second reason I divide up the time period is to leverage features of each period to craft nuanced tests that further help to distinguish between my claim and those of rival theories.

First, I conduct two types of tests to examine whether a similar relationship between vote concentration and turnout is apparent when analyses are limited to the five Lower House elections held between 1980 and 1993. To recap, in these elections, Japan was divided into between 129 and 130 multimember electoral districts, voters cast a single vote for a candidate, and the top n vote-getters in each district won a seat, where n ranged from two to six. Table 8.2 presents the same four specifications presented in Table 8.1, with the same set of controls but limits observations to elections in the 1980–1993 period. The coefficients on the indicators of voting concentration are positive and statistically significant in all specifications. Models 1 and 3, which use district-year fixed effects, tell us that municipalities that concentrated more votes on a single LDP winner have higher turnout than their same-district counterparts that concentrated fewer votes on a single LDP winner. Models 2 and 4, which use both district-year fixed effects and municipality fixed effects, tell us that when the same municipality concentrated more votes on a single LDP winner, its turnout rate was higher than when it concentrated fewer votes on a single LDP winner. Substantively, the coefficient on Single LDP Winner $VS_{m,t}$ in Model 1 tells us that on average, a one-standard-deviation increase in voting concentration is expected to translate into an 0.35% increase in turnout (mean turnout in this period was higher, at 79%).

Second, as I explained in Chapter 6, the 1980–1993 period was characterized by a very high degree of stability in municipality and electoral district boundaries. This means that Japanese municipalities faced the same set of competitors in the tournament, election after election. I can take advantage of this by conducting a first difference regression, which is an alternative means of estimating the effects of within-municipality changes in vote concentration. If, controlling for changes in other time-varying municipality-level features with consequences for turnout (such as population density), as well as features of a municipality's district-year

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

Table 8.2 *Municipalities that concentrate more votes on a single LDP winner have higher turnout in Lower House elections, 1980–1993.*

	Dependent variable: Turnout$_{m,t}$			
	(Model 1)	(Model 2)	(Model 3)	(Model 4)
Single LDP Winner VS$_{m,t}$	0.024***	0.020***		
	(0.002)	(0.003)		
Rank(Single LDP)			0.007***	0.004***
Winner VS$_{m,t}$)			(0.001)	(0.001)
Turnout$_{m,t-1}$	0.720***	−0.109***	0.724***	−0.112***
	(0.009)	(0.013)	(0.009)	(0.013)
Fiscal Strength$_{m,t}$	−0.008***	0.015***	−0.008***	0.015***
	(0.002)	(0.004)	(0.002)	(0.004)
Dependent Population$_{m,t}$	0.110***	0.052*	0.110***	0.057**
	(0.012)	(0.022)	(0.012)	(0.022)
Farming Population$_{m,t}$	0.033***	−0.050*	0.035***	−0.048
	(0.005)	(0.025)	(0.005)	(0.025)
Log(Population$_{m,t}$)	−0.008***	−0.085***	−0.007***	−0.086***
	(0.0004)	(0.011)	(0.0004)	(0.011)
Log(Per Capita Income$_{m,t}$)	−0.001	0.008	−0.001	0.007
	(0.002)	(0.005)	(0.002)	(0.005)
Population Density$_{m,t}$	−0.001***	0.021**	−0.001**	0.022**
	(0.0004)	(0.007)	(0.0004)	(0.007)
Observations	12,398	12,398	12,288	12,288
District-Year FE	Y	Y	Y	Y
Municipality FE	N	Y	N	Y
R^2	0.770	0.059	0.768	0.057

Robust standard errors clustered on municipality in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

that could influence turnout in all the municipalities therein, I find that increases in the degree to which a municipality concentrates votes on a single LDP winner are correlated with higher turnout, this will further bolster confidence in the theory.

Table 8.3 examines the effect of changes in vote concentration between two consecutive Lower House elections on change in turnout in those same elections for municipalities in the Lower House elections held between 1983 and 1993. Because my data begins in 1980, the 1983 election is the first election for which I can compute change variables (I call these Δ variables). In both specifications, the dependent variable is ΔTurnout$_{m,t}$, which is the municipality's turnout rate in the Lower House election held at t, minus its turnout in the previous Lower

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

Table 8.3 *Increases in voting concentration between two consecutive elections in the same municipality are associated with increases in turnout, 1980–1993.*

	Dependent variable: ΔTurnout$_{m,t}$	
	(Model 1)	(Model 2)
Δ Single LDP Winner VS$_{m,t}$	0.017***	
	(0.003)	
Δ Rank(Single LDP Winner VS$_{m,t}$)		0.003***
		(0.001)
Δ Fiscal Strength$_{m,t}$	0.012**	0.012**
	(0.004)	(0.004)
Δ Dependent Population$_{m,t}$	0.007	0.001
	(0.029)	(0.029)
Δ Farming Population$_{m,t}$	−0.033	−0.031
	(0.026)	(0.026)
Δ Log(Population$_{m,t}$)	−0.154***	−0.153***
	(0.032)	(0.033)
Δ Log(Per Capita Income$_{m,t}$)	0.018***	0.019***
	(0.005)	(0.005)
Δ Population Density$_{m,t}$	0.021	0.019
	(0.013)	(0.014)
Observations	12,393	12,224
District-Year FE	Y	Y
Municipality FE	Y	Y
R^2	0.019	0.017

Robust standard errors clustered on municipality in parentheses.
$^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

House election. Positive values on ΔTurnout$_{m,t}$ mean that municipality m increased its turnout between the two elections. In Model 1, I examine the effect of ΔSingle LDP Winner VS$_{m,t}$, which is the municipality's value on Single LDP Winner VS$_{m,t}$ in the Lower House election, minus its value on Single LDP Winner VS$_{m,t}$ in the previous Lower House election. Positive values on ΔSingle LDP Winner VS$_{m,t}$ mean that municipality m had higher values on Single LDP Winner VS$_{m,t}$ in the second election. In Model 2, I examine the effect of Δ Rank(Single LDP Winner VS$_{m,t}$), which is constructed analogously.

In both specifications, I control for changes in my six municipality-level variables, constructing analogous Δ variables for each. Both specifications also use municipality fixed effects, which control for

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

time-invariant features of municipality *m* that could plausibly be correlated with both changes in voting concentration and changes in turnout. Both specifications also use district-year fixed effects, which control for the possibility that systematic differences between districts in the second election could influence the degree to which voting concentration and turnout changed for all municipalities in that district. Because almost all municipalities are located in the same electoral district in the period, any changes to district-level features with consequences for turnout, such as the entry of new candidates projected to shake up the race, will act on all municipalities in that district simultaneously. This means that they are accounted for with the district-year fixed effect.

The coefficients on ΔSingle LDP Winner $VS_{m,t}$ and Δ Rank(SingleLDP Winner $VS_{m,t}$) are positive and statistically significant. Thus, increases in the degree to which the same municipality concentrated votes on a single LDP winner were associated with higher turnout in the municipality, even controlling for changes in other features of the municipality or its electoral district that might matter for turnout.

8.2.2 *Competing Explanations and Placebo Tests*

As I explained, in the 1980–1993 period, candidates competed in MMDs. In almost every district, the LDP ran more than one candidate, and in the vast majority of districts, at least two of those LDP candidates were successful. I can leverage these features of the period to design four placebo tests.

To recap, the theory expects that within district-years, the relationship between concentrating votes on a single LDP winner and turnout in Japanese municipalities will be positive. The mechanism thought to produce this relationship is the large prize that municipalities become eligible for as more and more of their voters cast their votes for the same LDP winner. As a municipality becomes more dedicated to the election of a single LDP winner, voters will become more motivated to set aside their true feelings about the incumbent (and ignore the fact that their vote is unlikely to make a difference to the outcome of the election) and make the effort to vote. In doing so, they are trying to avoid being the reason their municipality misses out on a large prize. If this is the case, then it follows that within district-years, municipalities that concentrate larger shares of votes cast on a *single* LDP winner will have higher turnout (as I show above), but municipalities that concentrate larger shares of votes on the LDP per se, whether this means delivering more votes to LDP winners (plural) or to LDP candidates (both winners and losers), will not necessarily have higher turnout. Put differently, a municipality that delivers more

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

votes to the LDP *could* have higher turnout, but only if those votes are *concentrated* on a single LDP winner.[14]

To examine these observable implications of the theory, I constructed:

$$\text{All LDP Winners VS}_{m,t} = \frac{\sum_{c \in m}\{LDP_{c,t} w_{c,t} v_{c,m,t}\}}{v_{m,t}}. \tag{8.3}$$

In words, this takes the number of votes received by all LDP winners in municipality m at time t, takes the sum of these, and divides this by the total number of valid votes cast in the municipality. To obtain a high value on All LDP Winners $\text{VS}_{m,t}$, municipalities must send more of the votes cast to LDP winners; however the distribution of votes among those LDP winners is arranged. In addition, I constructed:

$$\text{All LDP Candidates VS}_{m,t} = \frac{\sum_{c \in m}\{LDP_{c,t} v_{c,m,t}\}}{v_{m,t}}. \tag{8.4}$$

In words, this takes the number of votes received by all LDP candidates in municipality m at time t, takes the sum of these, and divides this by the total number of valid votes cast in the municipality. To obtain a high value on All LDP Candidates $\text{VS}_{m,t}$, municipalities must cast more votes for LDP candidates, regardless of whether those candidates win or lose and however the distribution of votes among them is arranged.

Table 8.4 presents two-way fixed effect regression models of $\text{Turnout}_{m,t}$ in the five Lower House elections held between 1980 and 1993. All specifications control for time-varying municipality-level characteristics that might matter for turnout, including turnout in the previous election, as well as district-year and municipality fixed effects. In Model 1, the independent variable of interest is All LDP Winners $\text{VS}_{m,t}$. Its coefficient is positive but not statistically significant. Thus, there is no evidence that municipalities delivering larger shares of votes cast to LDP winners per se had higher turnout than municipalities delivering smaller shares of votes cast to these winners. Whereas Table 8.2 showed that concentrating votes on a single LDP winner produces higher turnout, Model 1 in Table 8.4 shows that sending more votes to LDP winners, *plural*, is not associated with higher turnout.

In Model 2, I examine the effect of All LDP Candidates $\text{VS}_{m,t}$. Its coefficient is negative and not statistically significant. Thus, municipalities

[14] In Chapter 6, my analyses relied on variables capturing levels of electoral support delivered to both a single LDP winner and all LDP winners. Both variables were found to have a positive, statistically significant relationship with postelection transfers. If most Japanese municipalities concentrate votes on a single LDP winner, as they are incentivized to do under the tournament, their values on the two support variables might not be sufficiently different to pick up differences in transfers, especially since transfers are not solely the prerogative of LDP politicians (they also involve bureaucrats).

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

Table 8.4 *Municipalities delivering more votes to the LDP per se, whether to all LDP winners (Model 1) or to all LDP candidates, winning and losing (Model 2), do not have higher turnout in Lower House elections, 1980–1993. Similarly, municipalities that concentrate more votes on a single LDP loser (Model 3) and municipalities that concentrate more votes on a non-LDP winner (Model 4) do not have higher turnout.*

	Dependent variable: $\text{Turnout}_{m,t}$			
	(Model 1)	(Model 2)	(Model 3)	(Model 4)
All LDP Winners $\text{VS}_{m,t}$	0.005			
	(0.003)			
All LDP Candidates $\text{VS}_{m,t}$		−0.001		
		(0.004)		
Single LDP Loser $\text{VS}_{m,t}$			−0.009**	
			(0.003)	
Single Non-LDP+ Winner $\text{VS}_{m,t}$				−0.006
				(0.004)
$\text{Turnout}_{m,t-1}$	−0.106***	−0.105***	−0.103***	−0.105***
	(0.013)	(0.013)	(0.013)	(0.013)
$\text{Fiscal Strength}_{m,t}$	0.014***	0.014***	0.014***	0.014***
	(0.004)	(0.004)	(0.004)	(0.004)
$\text{Dependent Population}_{m,t}$	0.052*	0.051*	0.051*	0.052*
	(0.022)	(0.022)	(0.022)	(0.022)
$\text{Farming Population}_{m,t}$	−0.051*	−0.053*	−0.051*	−0.054*
	(0.025)	(0.025)	(0.025)	(0.025)
$\text{Log(Population}_{m,t})$	−0.086***	−0.086***	−0.086***	−0.087***
	(0.011)	(0.011)	(0.011)	(0.011)
$\text{Log(Per Capita Income}_{m,t})$	0.008	0.007	0.008	0.008
	(0.005)	(0.005)	(0.005)	(0.005)
$\text{Population Density}_{m,t}$	0.021**	0.022***	0.021**	0.022***
	(0.007)	(0.007)	(0.007)	(0.007)
Observations	12,398	12,398	12,398	12,398
District-Year FE	Y	Y	Y	Y
Municipality FE	Y	Y	Y	Y
R^2	0.053	0.053	0.054	0.053

Robust standard errors clustered on municipality in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

that delivered larger shares of votes cast to LDP candidates, both winning and losing, do not have systematically higher turnout than municipalities in the same district-year that delivered smaller shares of votes cast to LDP candidates. This is line with my expectations and provides further support

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

for my claim that it is support for a single LDP winner, not support for the LDP per se.

Next, the theory leads us to expect that municipalities that concentrate larger shares of votes on a single LDP *winner* will have higher turnout (as I have shown), but municipalities that concentrate larger shares of votes on a single LDP candidate who *loses* the election are unlikely to exhibit higher turnout. Why? To realize the kind of intense competition between municipalities over which will emerge as the most supportive, which is what encourages more and more voters in highly supportive municipalities to turn out, a candidate must be able to make her promises to influence central government bureaucrats credible. Any question mark over her ability to win the election reduces the credibility of those promises, which harms her ability to create this competition. In support of this, the analyses in Chapter 6 revealed that NTD allocations were not a function of how much municipalities supported their LDP candidate when those candidates lost the election.

To examine whether it is LDP candidates who win the election who are able to transform voting into a competition, and not those who lose, I constructed:

$$\text{Single LDP Loser VS}_{m,t} = \frac{max_{c \in m}\{LDP_{c,t}(1 - w_{c,t})v_{c,m,t}\}}{v_{m,t}}. \qquad (8.5)$$

This takes the number of votes received by losing LDP candidates in the municipality at time t, identifies the losing LDP candidate who won the largest number of votes in the municipality, and divides the number of votes won by this candidate by the total number of votes cast in the municipality. To obtain a high value on Single LDP Loser $\text{VS}_{m,t}$, municipalities have to cast more of their votes, however many are cast, for a single LDP loser.

In Table 8.4's Model 3, I examine the effect of Single LDP Loser $\text{VS}_{m,t}$. Its coefficient is negative and statistically significant, in line with my expectations. This means that municipalities that concentrate larger shares of votes on a single LDP loser tend to have systematically *lower* turnout than same-district municipalities that concentrate smaller shares of votes on a single LDP loser. This is evidence that LDP candidates who lose the election are unable to create the kind of competition among supportive municipalities that translates into higher turnout. Indeed, the fact that municipalities concentrating larger shares of votes on a single LDP loser appear to have *lower* turnout makes sense because municipalities concentrating smaller shares of votes on an LDP candidate who loses are likely concentrating larger shares of votes on an LDP candidate who wins, and I expect that those municipalities experience higher turnout.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

Finally, the evidence thus far suggests that tournaments are the preserve of LDP winners, not winners from the opposition parties. Chapter 6 confirmed that opposition winners do not seem able to direct money to supportive municipalities, even when they constituted the government after the 2009 election. Thus, we are unlikely to observe municipalities concentrating larger shares of votes cast on *non*-LDP winners exhibiting systematically higher turnout. However, I need to also be mindful of the finding in Chapter 6 that LDP-aligned *independent* winners, who were admitted into the LDP after the election, were granted access to central government resources after their victory and engaged in GBC in the exact same manner as ordinary LDP winners (who had gained the party's nomination prior to the election). In that chapter, I constructed $\text{LDP+}_{c,t}$ to indicate whether candidate c was either a member of the LDP at the time of the election or a conservative-aligned independent who joined the LDP after the election. Using this same variable, I constructed:

$$\text{Single Non-LDP+ Winner VS}_{m,t} = \frac{max_{c \in m}\{(1 - LDP+_{c,t})w_{c,t}v_{c,m,t}\}}{v_{m,t}}. \tag{8.6}$$

This takes the number of votes received by candidates who were *neither* an LDP winner *nor* an independent winner admitted into the LDP after the election in the municipality at time t, identifies the (non-LDP+) winner who won the largest number of votes, and divides the number of votes won by this candidate by the total number of votes cast in the municipality. To obtain a high value on Single Non-LDP+ Winner $\text{VS}_{m,t}$, municipalities have to cast more of their votes, however many are cast, for a single non-LDP (or LDP-aligned independent) winner.

In Table 8.4's Model 4, I examine the effect of Single Non-LDP+ Winner $\text{VS}_{m,t}$. Its coefficient is negative and not statistically significant. Thus, in line with my expectations, municipalities concentrating larger shares of votes on a non-LDP winner did not have higher turnout. This is further evidence that opposition winners do not engage in GBC.

8.2.3 After Electoral Reform, 1996–2014

Next, I turn to the 1996–2014 period. In this period (and since), Japan has been using a MMM system, in which a portion of candidates are elected via first-past-the-post in single-seat districts and another portion are elected from party lists in regional blocs via closed-list proportional representation. As Chapter 5 explains, whereas 91% of electoral districts were tournament-possible under the preelectoral reform period, this dropped to around 73% in the 1996, 2000, and 2003 elections,

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

around 62% in the 2005 election, and then around 50% in the 2009, 2012, and 2014 elections. Thus, the analyses in this section pertains to a smaller share of electoral districts.

As I explained in Chapter 6, the post-1994 period was characterized by instability in the boundaries of both municipalities and electoral districts. By 2005, 61% of the municipalities in existence in 2000 had undergone an amalgamation that had resulted in profound changes to its borders. In addition, two redistricting plans, one carried out prior to the 2003 election and the other carried out prior to the 2014 election, resulted in a redrawing of the boundaries of numerous electoral districts. Thus, in addition to changes to their *own* borders, many municipalities found themselves in one electoral district in one election, with one set of competitors, and then in a different electoral district in the next election, with a different set of competitors. Even in the absence of changes to the borders of one's own electoral district, if some of the municipalities in that district had merged between two elections, the arrangement of voters into municipalities would have changed. Chapter 4 explained why this variable – the degree of asymmetry in municipality size – is expected to matter enormously for both the overall support an LDP incumbent can hope to muster in a district and the amount of resources needed to muster it. This was corroborated with the analyses in Chapter 7.

For my purpose, these boundary changes mean that in contrast to the 1980–1993 period, I do not have a set of municipalities whose borders remained the same and whose competitors in the tournament also remained the same for a period of consecutive elections. For this reason, I implement the same regressions as above but rely on the lagged dependent variable to control for the possibility that underlying differences between municipalities could be driving my results, not municipality fixed effects. Table 8.5 presents the results of fixed effect regressions of $\text{Turnout}_{m,t}$ in the seven Lower House elections held between 1996 and 2014 as a function of Single LDP Winner $\text{VS}_{m,t}$ (Model 1) and Rank(Single LDP Winner $\text{VS}_{m,t}$) (Model 3), respectively. In both specifications, I control for turnout in the previous Lower House election and same six time-varying municipality-level characteristics that may matter for vote concentration and turnout. I include district-year fixed effects and report robust standard errors clustered on municipality.

The coefficients on both indicators of vote concentration are positive and statistically significant. Thus, controlling for other features of municipalities with consequences for turnout, including turnout in the previous election, I find that municipalities that concentrated larger shares of votes

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

Table 8.5 *Municipalities that concentrate more votes on a single LDP winner have higher turnout in Lower House elections, 1996–2014.*

	Dependent variable: $\text{Turnout}_{m,t}$	
	(1)	(2)
Single LDP Winner $\text{VS}_{m,t}$	0.033***	
	(0.005)	
Rank(Single LDP Winner $\text{VS}_{m,t}$)		0.005***
		(0.001)
$\text{Turnout}_{m,t-1}$	0.728***	0.733***
	(0.011)	(0.012)
Fiscal $\text{Strength}_{m,t}$	−0.008***	−0.008***
	(0.002)	(0.002)
Dependent $\text{Population}_{m,t}$	0.189***	0.213***
	(0.015)	(0.015)
Farming $\text{Population}_{m,t}$	0.004	0.009
	(0.007)	(0.008)
Log($\text{Population}_{m,t}$)	−0.008***	−0.009***
	(0.0004)	(0.001)
Log(Per Capita $\text{Income}_{m,t}$)	0.024***	0.025***
	(0.003)	(0.003)
Population $\text{Density}_{m,t}$	0.001***	0.002***
	(0.0003)	(0.0004)
Observations	12,538	8,597
District-Year FE	Y	Y
R^2	0.781	0.791

Robust standard errors clustered on municipality in parentheses.
***$p < 0.001$.
Regression models estimated with R's plm() package.

on a single LDP winner tended to have higher turnout than their same-district counterparts that concentrated smaller shares of votes on a single LDP winner. Substantively, the coefficient on Single LDP Winner $\text{VS}_{m,t}$ in Model 1 tells us that a one-standard-deviation increase in voting concentration is expected to translate into a 1% increase in turnout. The sample mean of $\text{Turnout}_{m,t}$ in this period was 0.66, meaning that on average, 66% of eligible voters in a municipality turned out to vote. Again, this is the *average* effect of increases in voting concentration; I expect increases in voting concentration to produce even larger increases in turnout among municipalities already concentrating many votes on LDP winners (examined below).

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

8.2.4 Competing Explanations and Placebo Tests

As I explained above, the theory expects that LDP candidates who win the election would have been able to create the perception that resource allocations, at least among highly supportive municipalities, are poised on a knife edge. It is this *perception*, that one's vote could determine whether one's municipality wins a large prize or has to settle for a smaller one, that drives up turnout. It is much less likely, in contrast, that LDP candidates who lose the election and winning candidates from other parties would have been able to create a similar perception. In support of this, my analyses in Chapter 6 found that winning LDP candidates were able to lean on central government bureaucrats in ways that translated into resources being a function of support, but neither their losing LDP counterparts nor their winning non-LDP counterparts were able to do so. Thus, the theory expects that within electoral districts, municipalities concentrating more votes on a single LDP winner will have higher turnout (as I have shown), but there will be no such relationship between concentrating votes on losing LDP candidates and concentrating votes on non-LDP winners, respectively, and turnout.

In Table 8.6, I present fixed effect regression models of $\text{Turnout}_{m,t}$ in the seven Lower House elections held between 1996 and 2014. In all specifications, I control for turnout in the previous Lower House election and same six time-varying municipality-level characteristics that I think might matter for both vote concentration and turnout. I include district-year fixed effects and report robust standard errors clustered on municipality. In Model 1, I limit observations to municipalities in the 458 district-years without LDP winners (defined as districts in which the LDP candidate did not place first). My variable of interest is Single LDP Loser $\text{VS}_{m,t}$, which is the number of votes cast for the losing LDP candidate, divided by the total number of valid votes cast in the municipality. This variable captures the degree to which municipalities concentrated their votes on the district's losing LDP candidate. Its coefficient is negative and not statistically significant. Thus, concentrating votes on an LDP candidate who loses the election is not associated with higher turnout.

In Model 2, I limit observations to municipalities in the 145 district-years in which the LDP candidate lost the electoral district but managed to enter parliament via PR. As I explained in Chapter 6, this set of LDP candidates lost the election by relatively narrow margins. Of all losing LDP candidates, these candidates were the ones who came closest to winning their districts. My analyses in Chapter 6 suggested that in these districts, postelection resource allocations were not a function of electoral support. Consistent with this, the coefficient on Single LDP Loser $\text{VS}_{m,t}$ in

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

Table 8.6 *Municipalities concentrating more votes on losing LDP candidates (Models 1 and 2) and DPJ winners (Model 3) do not have higher turnout in Lower House elections, 1996–2014.*

	Dependent variable: $\text{Turnout}_{m,t}$		
	(Model 1) (LDP Candidate Loses SSD)	(Model 2) (LDP Candidate Loses SSD, Wins PR Seat)	(Model 3) (DPJ Candidate Wins SSD)
Single LDP Loser $\text{VS}_{m,t}$	−0.014	−0.008	
	(0.009)	(0.014)	
Single DPJ Winner $\text{VS}_{m,t}$			0.004
			(0.011)
$\text{Turnout}_{m,t-1}$	0.724***	0.683***	0.722***
	(0.020)	(0.044)	(0.025)
Fiscal $\text{Strength}_{m,t}$	−0.007**	−0.009	−0.002
	(0.002)	(0.005)	(0.003)
Dependent $\text{Population}_{m,t}$	0.141***	0.172***	0.078*
	(0.028)	(0.052)	(0.030)
Farming $\text{Population}_{m,t}$	0.004	0.008	0.009
	(0.013)	(0.021)	(0.015)
$\text{Log}(\text{Population}_{m,t})$	−0.008***	−0.009***	−0.005***
	(0.001)	(0.002)	(0.001)
$\text{Log}(\text{Per Capita Income}_{m,t})$	0.021***	0.018*	0.015*
	(0.006)	(0.009)	(0.008)
Population $\text{Density}_{m,t}$	0.001*	0.001*	−0.0003
	(0.0004)	(0.001)	(0.0003)
Observations	3,941	1,199	2,091
District-Year FE	Y	Y	Y
R^2	0.756	0.768	0.785

Robust standard errors clustered on municipality in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

Model 2 is similarly negative and not statistically significant. The absence of a relationship between vote concentration and turnout in municipalities in these districts is consistent with the theory's expectation that losing LDP candidates find it difficult to create the perception in supportive municipalities that every vote counts, due to its potential to bring about a sizeable increase in its resource allocation.

In Model 3, I limit observations to municipalities in the 280 district-years with Democratic Party of Japan (DPJ) winners (defined as districts

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

in which the DPJ candidate placed first). Let $DPJ_{c,t}$ indicate whether candidate c was a member of DPJ at time t. Using a similar equation as above, I constructed:

$$\text{Single DPJ Winner VS}_{m,t} = \frac{max_{c \in m}\{DPJ_{c,t} w_{c,t} v_{c,m,t}\}}{v_{m,t}}. \qquad (8.7)$$

In words, this takes the number of votes received by the DPJ winner in municipality m at time t, and divides this by the number of valid votes cast in the municipality. This variable captures the degree to which municipalities concentrated their votes on the district's winning DPJ candidate. Its coefficient is positive but not statistically significant. This is further evidence that winning DPJ candidates did not contest elections by dangling the promise of ever-larger resource allocations over the heads of voters in supportive municipalities, which is the mechanism that produces the positive relationship between concentrating votes on district winners and turnout in LDP-won districts.

8.3 A CONVEX RELATIONSHIP BETWEEN VOTE CONCENTRATION AND TURNOUT

Next, we move to Hypothesis 6. Here, I am interested in whether, within electoral districts, the relationship between concentrating votes on a single LDP winner and turnout is *convex*, rather than linear. The theory holds that LDP politicians can amass enough votes to win by pitting municipalities against each other in a competition for resources. By offering the largest prize to the most supportive municipality, the second-largest prize to the second-most supportive municipality, and so on, and making the difference in amounts received larger at higher ranks, politicians can create a situation where handfuls of votes in relatively supportive municipalities can mean the difference between a large prize and a much smaller one. This gives voters in those municipalities incentives to set aside their personal feelings about the incumbent and think primarily about the degree of influence their vote wields over the size of their municipality's prize.

In Chapter 6, I found that the distribution of postelection resource allocations within electoral districts exhibited this pattern: Namely, further increases in electoral support among municipalities that were already very supportive yielded much a larger increase in postelection resource allocations than a similar increase in electoral support among municipalities that were less supportive. I expect to observe a similar relationship between electoral support and turnout: Namely, I expect that turnout will rise at an increasing rate as municipalities concentrate more and more

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

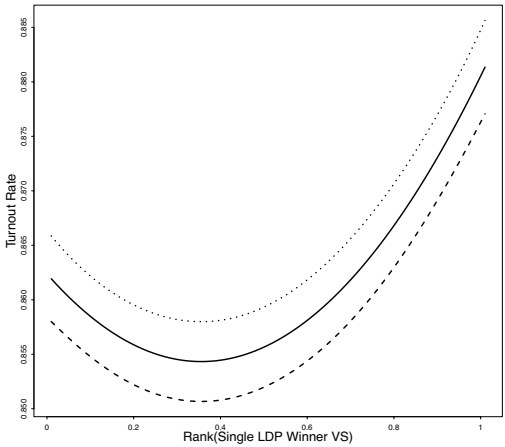

Figure 8.1 The relationship between rank and turnout is convex, 1980–2014. Predicted values (with 95% confidence intervals) from a regression of turnout in Japanese municipalities in the twelve Lower House elections held between 1980 and 2014 on a cubic specification of Rank(Single LDP Winner $VS_{m,t}$), municipality-level controls, and district-year fixed effects.

votes on a single LDP winner. Put differently, a municipality experiencing a one-unit increase in vote concentration at the top of the ranking will experience a larger increase in turnout than a municipality experiencing the same one-unit increase in vote concentration at the middle or bottom of the ranking. This would be evidence that decisions to vote are being influenced by the size of the prize one's group stands to win.

In Table A.16, I present the result of a fixed effect regression of $Turnout_{m,t}$ as a function of Rank(Single LDP Winner $VS_{m,t}$), as well as its quadratic and cubic transformations, for all municipalities in the twelve Lower House elections held between 1980 and 2014. As I explained in Chapter 6, a joint hypothesis test can tell us whether the coefficients on my measure of vote concentration, as well as its quadratic and cubic transformations, are jointly significant. The model controls for time-varying municipality-level characteristics that matter for turnout, as well as district-year fixed effects. I report robust standard errors clustered on municipality.

The joint hypothesis test is statistically significant. This is evidence that the relationship between concentrating votes on a single LDP winner and turnout is not linear. To visually depict the convexity, Figure 8.1 plots predicted values and 95% confidence intervals from the regression model in Table A.16. The figure shows that once a municipality is at the median or above in terms of vote concentration (0.5–1 on the x-axis), the effects of further increases in rank (vote concentration) increase at an increasing rate. For municipalities at the very top of the ranking, the

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

returns to further increases in vote concentration are the largest. Substantively, a municipality that increases its Rank(Single LDP Winner $VS_{m,t}$) from 0.95 to 1, for example, can expect an increase in turnout of 0.40%. The same increase in rank translates into smaller increases in turnout among municipalities at lower ranks. A municipality that increases its Rank(Single LDP Winner $VS_{m,t}$) from 0.8 to 0.85 can expect an increase in turnout of 0.30%, while a municipality that increases its Rank(Single LDP Winner $VS_{m,t}$) from 0.5 to 0.55 can only expect an increase in turnout of 0.11%.[15]

8.4 IMPACT OF COMPETITIVENESS DEPENDS ON VOTE CONCENTRATION

While the above tests held all variables operating at the level of the district-year constant and examined the relationship between support for a single LDP winner and turnout among municipalities in the same district-year, we can also harness the variation at this level to devise further tests of Hypothesis 5.

As I mentioned above, a large literature has established that more voters go the polls when a race is projected to be close. One reason for this is that when a race is close, a single voter stands more of a chance of influencing who wins. For incumbents seeking reelection, the fact that many voters lack a compelling reason to make the effort to turn out unless they expect the race to be close presents a problem, the severity of which only worsens as incumbents gain in experience and electoral security. The problem facing incumbents is: How can they motivate enough people to turn out and vote for them, election after election, even with the knowledge that their vote wields little-to-no influence over who wins? GBC more generally, and the form it takes with a dominant party, a tournament, presents a solution to this problem. By making the allocation of government resources a function of electoral support and calibrating the amounts received such that the most supportive municipalities stand to receive significantly more money with only a few more votes, incumbents are able to increase the amount of influence wielded by a single vote. Individual votes continue to wield little-to-no influence over *who wins the election*, but individual votes in supportive municipalities now wield an extraordinary degree of influence on another dimension: the amount of *resources* one's municipality receives.

[15] The same convex relationship is visible when observations are limited to the preelectoral and postelectoral reform periods.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

This leads to the expectation that, across electoral districts, turnout will be higher in races projected to be competitive. But within electoral districts, the tournament theory leads us to expect that projections of competitiveness will have a *heterogeneous* impact on turnout rates, depending on where in the ranking a given municipality is likely to end up. In municipalities that are highly supportive of the incumbent, the competition for resources has magnified the influence of a single vote and decisions to turn out are primarily motivated by attempts to influence the size of the municipality's prize. As such, projections of increased competitiveness are expected to exert a smaller impact on turnout. In municipalities that are less-supportive of the incumbent, in contrast, the influence of a single vote has not been magnified in the same way. In these municipalities, decisions to turn out will be less motivated by the desire to influence the amount of resources received by one's municipality. As such, projections of closeness are likely to exert a larger impact on turnout.

To examine this, it would be ideal to have measures of district-level competitiveness derived in advance of elections, which reflected perceptions in the electorate to how competitive the upcoming race was going to be. I could then see whether, in electoral districts where races were projected to be competitive, turnout increased *less* in municipalities that are highly supportive of the incumbent than in municipalities that are less supportive. In lieu of an independently derived measure, I compute the variable $\text{Margin}_{d,t}$. As I explained above, this is the margin of victory of the "last" winner in a district-year.[16] Critically, I use $\text{Margin}_{d,t}$ as a proxy for how competitive voters *anticipated* that the race would be in their district. Thus, if a district has a large $\text{Margin}_{d,t}$ (indicating a *less* competitive election), I assume that voters knew the race would be less competitive when they made their decisions to vote. Similarly, if a district has a small $\text{Margin}_{d,t}$ (indicating a *more* competitive election), I assume that voters knew the race would be competitive when they made their decisions to vote.[17]

In Table 8.7, I present the results of two-way fixed effect regressions of Japanese municipalities in the twelve Lower House elections held between 1980 and 2014. In both models, the dependent variable is turnout in municipality m ($\text{Turnout}_{m,t}$). In Model 1, my independent variable of interest is $\text{Margin}_{d,t}$, which is measured at the level of the district-year and is therefore identical for all municipalities in a given district-year.

[16] In an SSD, this is the difference in votes received by the winner and loser; in a multi-seat district, it is the difference in votes received by the nth-placed candidate and the $n + 1$-placed candidate, where n is district magnitude.

[17] Because $\text{Margin}_{d,t}$ is measured after votes have been cast, it is possible that voters anticipated a close election, but the outcome was not one, or vice versa. A stronger test of this hypotheses would use survey or other data to derive this measure.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

Higher values of $\text{Margin}_{d,t}$ indicate *less* competitive elections. In Model 2, I am interested in the effect of $\text{Margin}_{d,t}$, Rank(Single LDP Winner $\text{VS}_{m,t}$) (the independent variable from above), and their interaction. Both models control for the same time-varying municipality-level characteristics that might influence the relationship between vote concentration and turnout, including turnout in the previous Lower House election. Both models also use municipality fixed effects to control for the possibility that time-invariant features of municipalities could be driving any relationship I find, and year fixed effects to control for systematic features of a given election year that could influence turnout in all municipalities.

Importantly, the models in Table 8.7 are different from those presented earlier in this chapter in that they do not use district-year fixed effects. Above, I was interested in comparing municipalities in the same district-year. A district-year fixed effect enabled me to control for all the ways district-years differ from each other. Here, I am interested in leveraging variation that occurs at the level of the district-year (the competitiveness of the race). I want to compare municipalities in competitive races with otherwise-similar municipalities in less competitive races. My first question of interest is whether, controlling for other differences between municipalities, decreases in competitiveness (increases in margin) decrease turnout (Model 1). My second question of interest is whether, controlling for other differences between municipalities, decreases in competitiveness (increases in margin) decrease turnout *less* in highly supportive municipalities (Model 2).

Because I am not using district-year fixed effects, I need to control for other features of district-years that might exert a systematic effect on competitiveness, vote concentration, and turnout. In addition to year fixed effects, which also control for the fact that two different electoral systems were used during this time, I control for population per seat (the degree to which the district is malapportioned), the number of seats available (district magnitude), the number of candidates running in the district, and the number of LDP candidates running in the district. I also control for the degree of asymmetry in municipality size, which the previous chapter showed matters for the overall level of support LDP winners can hope to muster in a district.[18] I report robust standard errors clustered on the municipality.

In Model 1, the coefficient on $\text{Margin}_{d,t}$ is negative and statistically significant. This means that, controlling for time-invariant and time-varying features of municipalities, time-varying features of a municipality's district, and differences across elections, municipalities in districts in which

[18] District-level variables were described in Chapter 7.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

Table 8.7 *Increases in competitiveness impact turnout differently, depending on how supportive a municipality is for its LDP winner, 1980–2014.*

	Dependent variable: $\text{Turnout}_{m,t}$	
	(Model 1)	(Model 2)
$\text{Margin}_{d,t}$	−0.123***	−0.155***
	(0.003)	(0.005)
Rank(Single LDP Winner $\text{VS}_{m,t}$)		−0.002
		(0.001)
$\text{Margin}_{d,t}$:Rank(Single LDP Winner $\text{VS}_{m,t}$)		0.057***
		(0.008)
$\text{Turnout}_{m,t-1}$	0.271***	0.221***
	(0.009)	(0.010)
Fiscal $\text{Strength}_{m,t}$	0.005	0.005
	(0.004)	(0.004)
Dependent $\text{Population}_{m,t}$	0.296***	0.293***
	(0.017)	(0.019)
Farming $\text{Population}_{m,t}$	0.053*	0.046
	(0.022)	(0.025)
Log($\text{Population}_{m,t}$)	−0.021***	−0.034***
	(0.005)	(0.006)
Log(Per Capita $\text{Income}_{m,t}$)	0.019***	0.016**
	(0.005)	(0.005)
Population $\text{Density}_{m,t}$	0.039***	0.050***
	(0.006)	(0.007)
$\text{HI}_{d,t}$	0.011*	0.012
	(0.005)	(0.007)
Log(Number of $\text{Municipalities}_{d,t}$)	−0.004*	−0.005*
	(0.002)	(0.002)
People Per $\text{Seat}_{d,t}$	−0.002***	−0.004***
	(0.001)	(0.001)
Number LDP $\text{Candidates}_{d,t}$	0.008***	0.008***
	(0.001)	(0.001)
Number of $\text{Candidates}_{d,t}$	0.004***	0.005***
	(0.0003)	(0.0003)
Total $\text{Seats}_{d,t}$	−0.005***	−0.007***
	(0.001)	(0.001)
Observations	24,936	20,885
Municipality Fixed Effects	Y	Y
Year Fixed Effects	Y	Y
R^2	0.276	0.272

Increases in margin separating winner from loser (*decreases* in competitiveness) have a negative impact on turnout in municipalities in Lower House elections, 1980–2014 (Model 1). However, they have a *smaller* negative impact in municipalities positioned to do well in the tournament (those concentrating larger shares of the vote on a single LDP winner) (Model 2).

Robust standard errors clustered on municipality in parentheses.
$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

the margin separating the last winner and first loser is larger – indicating a *less* competitive election – have *lower* turnout. My use of municipality fixed effects means that I can interpret this as the effect of increases in margin (decreases in district-level competitiveness) within the same municipality over time. The negative, statistically significant coefficient on $\text{Margin}_{d,t}$ means that when the same municipality is in a district with a larger margin separating last winner from first loser (a *safer* district), its turnout rate is lower than when it is in a district with a smaller margin (a *competitive* district). Substantively, the coefficient on $\text{Margin}_{d,t}$ in Model 1 tells us that a one-standard-deviation increase in $\text{Margin}_{d,t}$ is expected to depress turnout by 1.80%. Thus, the competitiveness of races in Japan systematically affects turnout, as it does elsewhere.

In Model 2, I am interested in whether $\text{Margin}_{d,t}$ exerts a systematically *different* effect on turnout, depending on whether the municipality concentrates a large share of votes on a single LDP winner, which puts it in the running for a large prize after the election, or concentrates a small share of votes on a single LDP winner, which would put it out of contention for a large prize. The coefficient on $\text{Margin}_{d,t}$ remains negative and statistically significant, while the coefficient on $\text{Margin}_{d,t}*$ Rank(Single LDP Winner $\text{VS}_{m,t}$) is positive and statistically significant. This means that the effect of district-level competitiveness is, in fact, different, depending on the degree to which a municipality concentrates votes on a single LDP winner.

For municipalities whose Rank(Single LDP Winner $\text{VS}_{m,t}$) is 0, meaning they ranked last in their district in terms of identifying with a single LDP winner, the effect of a one-unit increase in $\text{Margin}_{d,t}$ on turnout is −0.155.[19] Substantively, for the least supportive municipality in the district, a one-standard-deviation increase in $\text{Margin}_{d,t}$ (a decrease in competitiveness) is expected to lower turnout by 2.33%. In contrast, for municipalities whose Rank(Single LDP Winner $\text{VS}_{m,t}$) is 1, meaning they ranked first in terms of identifying with a single LDP winner, the effect of a one-unit increase in $\text{Margin}_{d,t}$ on turnout is −0.098.[20] Substantively, for municipalities at the highest rank, a one-standard-deviation increase in $\text{Margin}_{d,t}$ (a decrease in competitiveness) is expected to lower turnout by 1.47%. Thus, while decreases in competitiveness have a negative effect on turnout in all municipalities, decreases in competitiveness have a *larger* negative effect on turnout in municipalities not slated to do well in the tournament.

Figure 8.2 plots the average marginal effect of $\text{Margin}_{d,t}$, with 95% confidence intervals, at different levels of Rank(Single LDP

[19] $-0.155 + (0.057 * 0) = -0.155$.
[20] $-0.155 + (0.057 * 1) = -0.098$.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

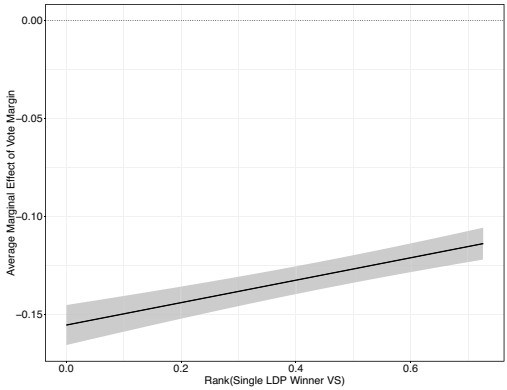

Figure 8.2 The black line depicts the average marginal effect of $\text{Margin}_{d,t}$, with 95% confidence intervals, at different levels of Rank(Single LDP Winner $\text{VS}_{m,t}$). Estimates are from Model 2 of Table 8.7.

Winner $\text{VS}_{m,t}$). Among the relatively unsupportive municipalities (lower values of Rank(Single LDP Winner $\text{VS}_{m,t}$)), the effect of increases in $\text{Margin}_{d,t}$ (decreases in competitiveness) on turnout is more negative than among the relatively supportive municipalities (higher values of Rank(Single LDP Winner $\text{VS}_{m,t}$)).[21]

It is worth pointing out that, for the most part, the direction and effects of the district-level controls is consistent with the existing literature. The coefficient on People Per Seat is negative and statistically significant. This indicates that, controlling for other differences between municipalities, both in their own characteristics and in characteristics of their districts, municipalities in districts with more people per representative (meaning larger electorate sizes) have lower turnout. The coefficient on Total Seats is also negative and statistically significant. This means that on average, municipalities in districts with larger district magnitude have lower turnout. The coefficients on Number of Candidates and Number of LDP Candidates are both positive and statistically significant, indicating that on average, municipalities in districts with more candidates running and districts with more LDP candidates running, respectively, have higher turnout. District-level asymmetry, my variable of interest in Chapter 7, is positively related to turnout in Model 1, but not in Model 2. In Chapter 7, we found that district-level asymmetry lowers support for the LDP. If this is the case, then these districts may produce more competitive races. To the extent they do so, any effects of this variable may be captured by $\text{Margin}_{d,t}$.

[21] The same relationship between variables is present when observations are limited to the preelectoral and postelectoral reform periods.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

8.4 Competitiveness Depends on Vote Concentration

8.4.1 Competing Explanations

In line with the theory, the above analyses demonstrated that the impact of district-level competitiveness on turnout is different at different levels of voting concentration. An intriguing finding about elections in Japan, at least under the old electoral system, is that competitiveness exerts a larger impact on turnout in rural areas, where social ties are denser (Cox, Rosenbluth and Thies 1998). The authors studied this in a regression with electoral districts as observations and interacted the competitiveness of the race in each district with measures of the electoral district's ruralness. This represents a confounder that I ought to take seriously: Could my finding that the impact of competitiveness is different at different levels of identification with a single LDP winner be explained by the fact that the impact of competitiveness is also different at different levels of ruralness?

In Table 8.8, I present the results of two-way fixed effect regression models of Japanese municipalities in the twelve Lower House elections held between 1980 and 2014. In both models, the dependent variable is $\text{Turnout}_{m,t}$. Both models control for the same time-varying municipality-level characteristics as above, including turnout in the previous election; the same time-varying district-level characteristics as above; and use year fixed effects and municipality fixed effects (also as above). I report robust standard errors clustered on the municipality.

In Model 1, I use an interaction between $\text{Margin}_{d,t}$ and Population Density to examine whether their findings hold in my data, in which observations are municipalities (not districts) in the entire period (not just prior to 1993). I find that the coefficient on $\text{Margin}_{d,t}$ is negative and statistically significant, while the coefficient on $\text{Margin}_{d,t}$*Population Density is positive and statistically significant. Thus, the impact of district-level competitiveness on turnout does appear to be *different*, depending on ruralness. Substantively, for the least dense municipality in my sample (with 1.3 people per square kilometer), a one-standard-deviation increase in $\text{Margin}_{d,t}$ (the district becomes less competitive) is estimated to decrease turnout by 1.88%. For the most dense municipality in my sample (19,275 people per square kilometer), the same one-standard-deviation increase in $\text{Margin}_{d,t}$ is estimated to decrease turnout by only 1.29%. Thus, while declines in competitiveness (increases in $\text{Margin}_{d,t}$ separating winner from loser) negatively impacts turnout everywhere, they have a larger negative effect in the most dense (rural) municipalities. This is what Cox, Rosenbluth and Thies (1998) found at district level.

In Model 2, I include both the interaction of interest from Table 8.7 (Model 1) which was $\text{Margin}_{d,t}$* Rank(Single LDP Winner $\text{VS}_{m,t}$),

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

Table 8.8 *Increases in margin separating winner from loser (decreases in competitiveness) have a larger negative impact on turnout in rural (less population dense) municipalities (Model 1). My finding that increases in margin have a smaller negative impact on turnout in municipalities slated to do well in the tournament is not explained by this (Model 2).*

	Dependent variable: $\text{Turnout}_{m,t}$	
	(Model 1)	(Model 2)
$\text{Margin}_{d,t}$	−0.129***	−0.166***
	(0.003)	(0.005)
$\text{Population Density}_{m,t}$	0.035***	0.044***
	(0.006)	(0.007)
Rank(Single LDP Winner $\text{VS}_{m,t}$)		−0.003*
		(0.001)
$\text{Margin}_{d,t}$:$\text{Population Density}_{m,t}$	0.011***	0.017***
	(0.002)	(0.004)
$\text{Margin}_{d,t}$:Rank(Single LDP Winner $\text{VS}_{m,t}$)		0.065***
		(0.008)
$\text{Turnout}_{m,t-1}$	0.271***	0.221***
	(0.009)	(0.010)
$\text{Fiscal Strength}_{m,t}$	0.006	0.007
	(0.004)	(0.004)
$\text{Dependent Population}_{m,t}$	0.293***	0.287***
	(0.016)	(0.019)
$\text{Farming Population}_{m,t}$	0.041	0.033
	(0.022)	(0.024)
Log($\text{Population}_{m,t}$)	−0.023***	−0.037***
	(0.005)	(0.006)
Log($\text{Per Capita Income}_{m,t}$)	0.021***	0.018***
	(0.005)	(0.005)
$\text{HI}_{d,t}$	0.011*	0.011
	(0.005)	(0.007)
Log($\text{Number of Municipalities}_{d,t}$)	−0.004*	−0.004*
	(0.002)	(0.002)
$\text{People Per Seat}_{d,t}$	−0.002***	−0.004***
	(0.001)	(0.001)
$\text{Number LDP Candidates}_{d,t}$	0.008***	0.007***
	(0.001)	(0.001)
$\text{Number of Candidates}_{d,t}$	0.004***	0.005***
	(0.0003)	(0.0003)
$\text{Total Seats}_{d,t}$	−0.006***	−0.007***
	(0.001)	(0.001)
Observations	24,936	20,885
Municipality Fixed Effects	Y	Y
Year Fixed Effects	Y	Y
R^2	0.278	0.274

Robust standard errors clustered on municipality in parentheses.
$^{**}p < 0.01$, $^{***}p < 0.001$.
Regression models estimated with R's plm() package.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

and the interaction of interest from Table 8.8 (Model 1) which was $\text{Margin}_{d,t}$*Population Density. The coefficient on $\text{Margin}_{d,t}$ remains negative and statistically significant, while the coefficient on $\text{Margin}_{d,t}$* Rank(Single LDPWinner $\text{VS}_{m,t}$) remains positive and statistically significant. For a municipality at the mean of population density (559 people per square kilometer), these coefficients tell us that the effect of a one-standard-deviation increase in $\text{Margin}_{d,t}$ on turnout in the municipality concentrating the least votes on a single LDP winner is −2.35%. In contrast, the effect of the same one-standard-deviation increase in $\text{Margin}_{d,t}$ in a municipality concentrating the most votes on a single LDP winner is 1.37%. The bottom line is, even if we control for the fact that the impact of competitiveness is conditional upon population density, with competitiveness increasing turnout more in rural municipalities than in urban areas, we still find that the impact of competitiveness is conditional upon the share of votes concentrated on a single LDP winner in the municipality. Municipalities ranked highest in support for their LDP incumbent are the least affected by increases or decreases in the competitiveness of the race.

8.5 TAKEAWAYS FOR JAPANESE POLITICS

We have found that turnout in Japanese elections is decisively influenced by the degree to which a voter's municipality identifies with a single LDP winner. Within electoral districts, as municipalities concentrate more votes on a single LDP winner, their turnout increases, with further increases in vote concentration among municipalities already concentrating large shares of their votes on a single LDP winner translating into even larger increases in turnout. The same relationship is not observed when municipalities concentrate more votes on a single winner from another party. Nor is it observed when municipalities concentrate more votes on a losing LDP candidate. Nor is it observed when municipalities spread their votes out across several LDP winners (possible in multiseat districts prior to 1994). The positive relationship between vote concentration and turnout in electoral districts with LDP winners is consistent with the tournament's expectation that many Japanese voters make their decisions to turn out with an eye to the amount of resources their municipality is slated to receive after the election.

This means that a portion of voters in Japan are making the effort to vote in order to receive an allocation of government resources that they anticipate their municipality might not receive if they did not make the effort to turn out. Whereas political scientists tend to believe that high turnout is indicative of a healthy democracy with an engaged populace,

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008

the findings in this chapter imply that high turnout may not be an unalloyed good. In some instances, it may be driven by a decidedly undemocratic competition for resources, in which voters are turning out and voting for their LDP incumbent not because they are partial to her policies or believe she is competent, but because they are unwilling to put their community's access to needed resources on the line. If the extent to which voters can turn their back on these incentives without material ramification is a function of how needy their community is, then the tournament creates a situation in which voters in communities that stand to benefit the *most* from well-structured, growth-oriented government policies are the ones with the least amount of voice. The ability of these voters to use elections to transmit their preferences to elected officials and hold those officials accountable for their actions has been curtailed.

Downloaded from https://www.cambridge.org/core. Oxford Union Society, on 11 May 2025 at 21:01:11, subject to the Cambridge Core terms of use, available at https://www.cambridge.org/core/terms. https://doi.org/10.1017/9781009588522.008